

ASSIGNMENT

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Course :M.Tech VLSI &ES

Q1. Simulate the I–V characteristics of PN junction diode using nanoHUB simulation tool.

Objectives:

- Study the I-V characteristics of Si PN diode.
- Study the I-V characteristics of Ge PN diode.
- Study the effect of doping concentrations of Si and Ge PN diodes.

Procedure 1:

Open a browser and go to **nanoHUB.org**.

Log in with your nanoHUB user ID.

In the search bar, type “**PN Junction Lab**” and open the tool.

I–V of Si diode

In the **Materials** tab, set **Material = Si (Silicon)**.

Go to the **Structure / Doping** panel at the bottom.

Under **P-type region**, set **acceptor concentration (Na)** value.

Under **N-type region**, set **donor concentration (Nd)** value.

$$Na = 1 \times 10^{17} \text{ cm}^{-3}, Nd = 1 \times 10^{16} \text{ cm}^{-3}$$

Keep **minority carrier lifetimes** at default values (1e-10 s for electrons and holes).

Simulate the device with the applied voltage of 1V under Environment section

At the top-right, choose **Result: I–V Characteristics** from the drop-down menu.

Click **Simulate**.

After the run completes, I-V curve appears on the screen

Procedure 2:

To change the material to Germanium

Under the materials section change the material to Germanium and plot the I-V characteristics .

Simulate the device with the applied voltage of 1V under Environment section

At the top-right, choose **Result: I–V Characteristics** from the drop-down menu.

Click **Simulate**.

After the run completes, the I-V curve appears on the screen.

Procedure 3:

The doping concentrations can be changed by clicking the value of the concentration.

$$\text{Case 1 (low doping): } Na = 1 \times 10^{16} \text{ cm}^{-3}, Nd = 1 \times 10^{15} \text{ cm}^{-3}$$

$$\text{Case 2 (medium doping): } Na = 1 \times 10^{17} \text{ cm}^{-3}, Nd = 1 \times 10^{16} \text{ cm}^{-3}$$

$$\text{Case 3 (high doping): } Na = 1 \times 10^{18} \text{ cm}^{-3}, Nd = 1 \times 10^{17} \text{ cm}^{-3}$$

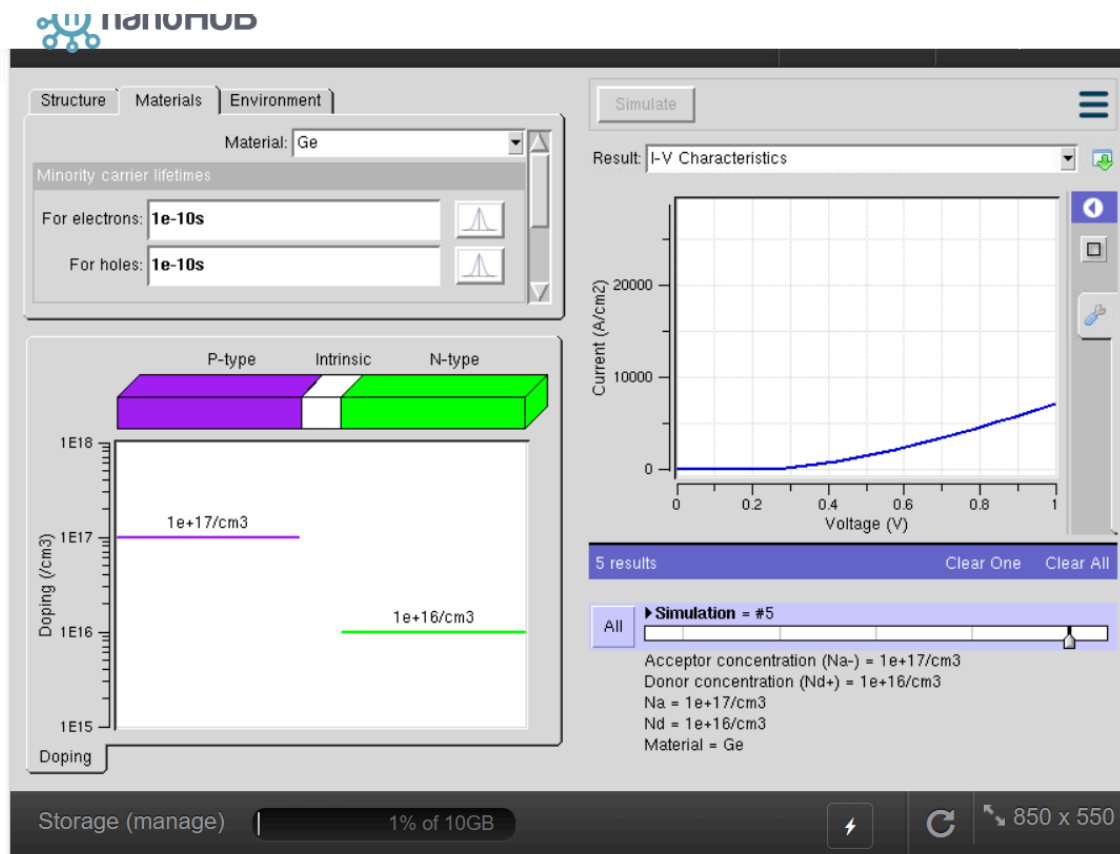
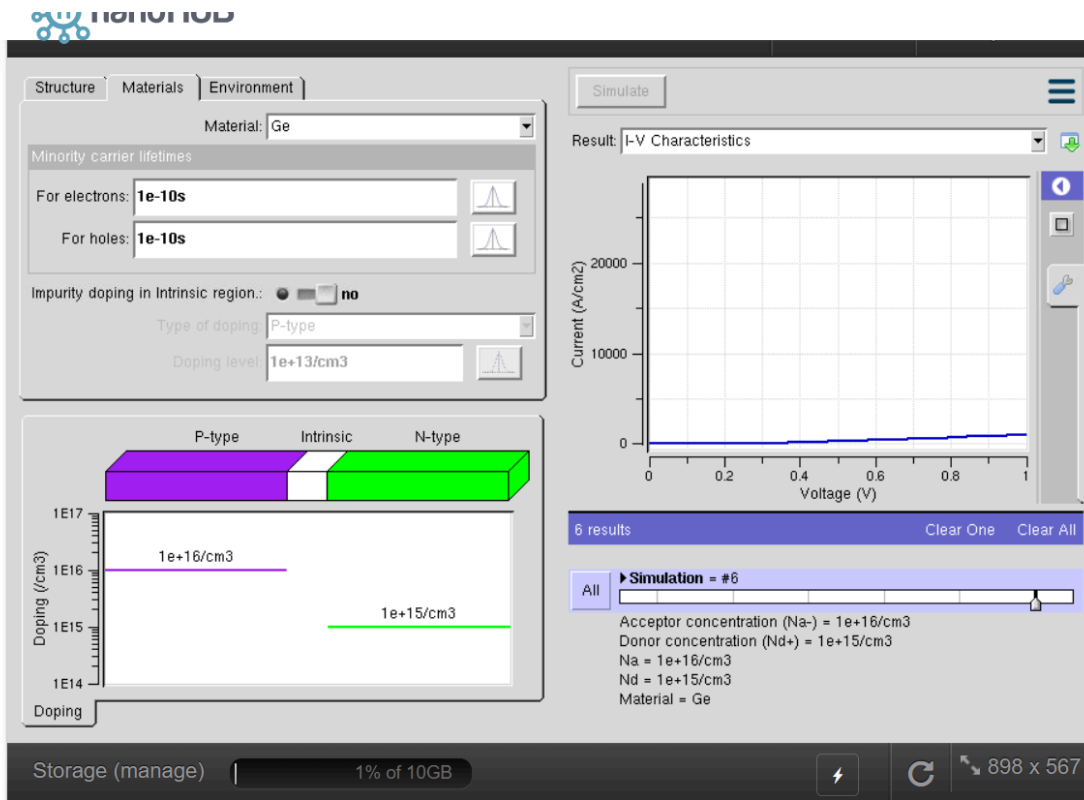
Simulate the device with the applied voltage of 1V under Environment section

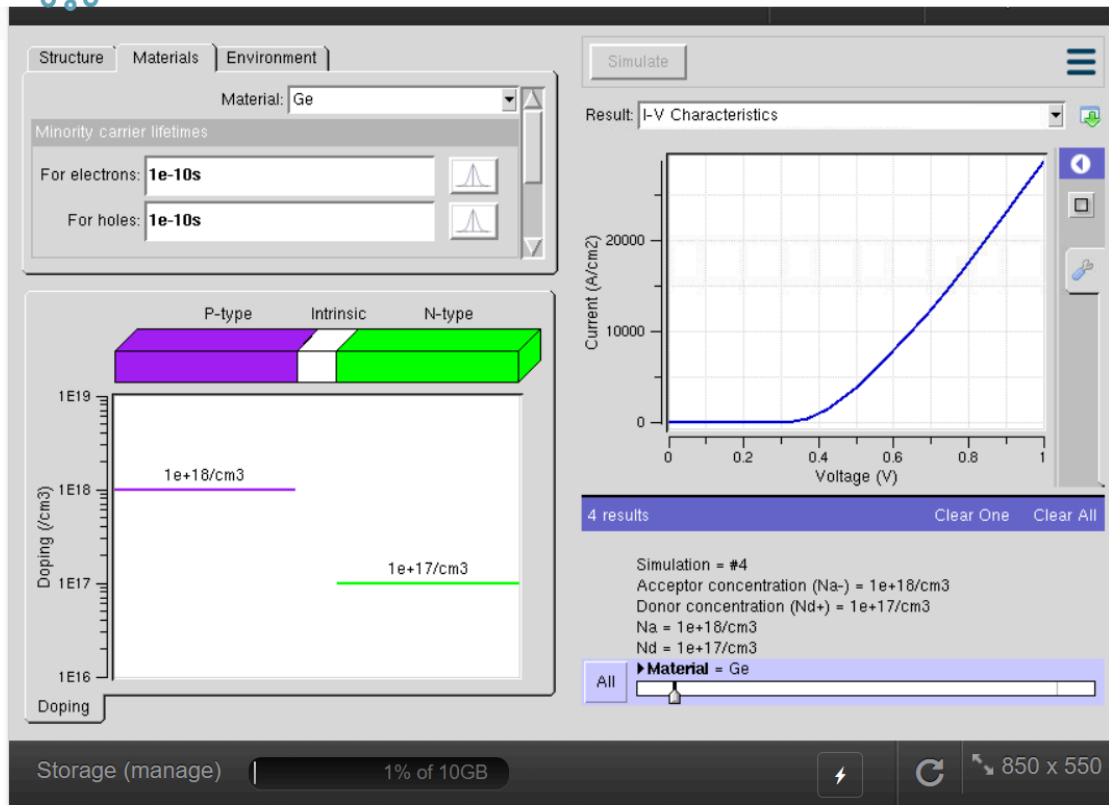
At the top-right, choose **Result: I–V Characteristics** from the drop-down menu.

Click **Simulate**.

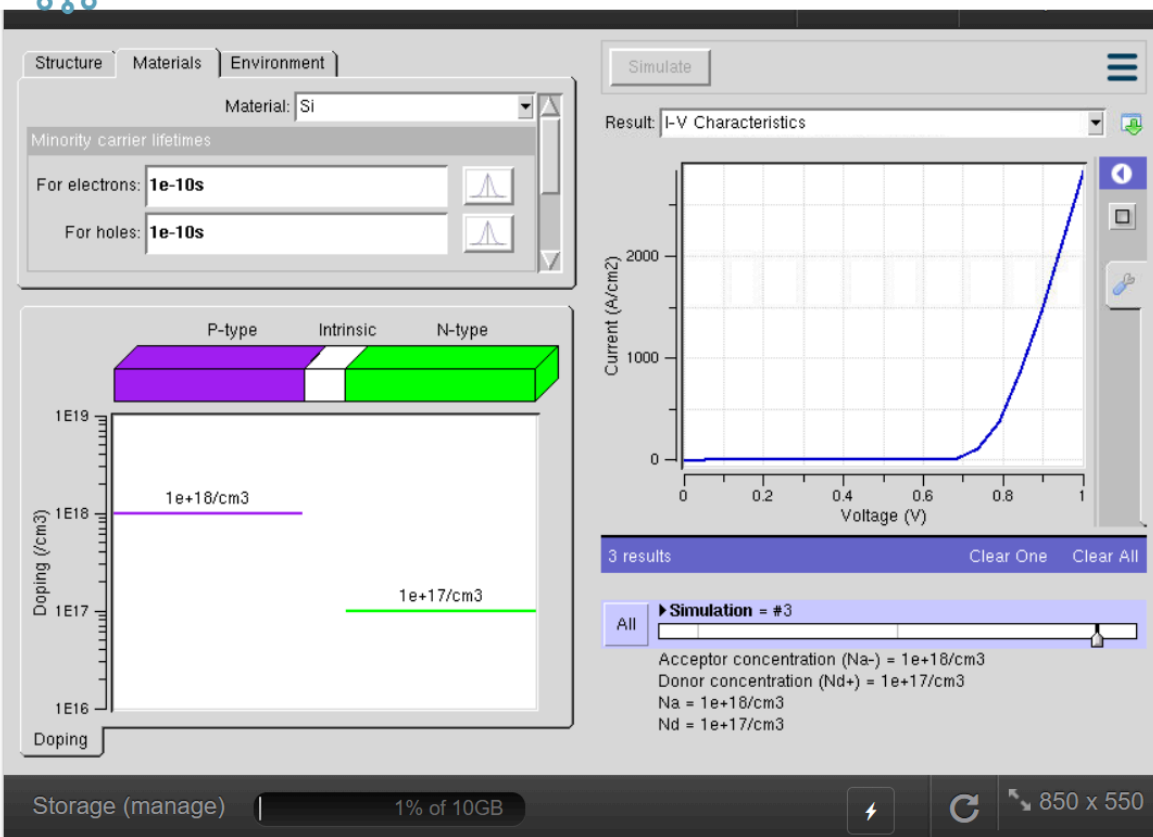
After the run completes, I-V curve appears on the screen.

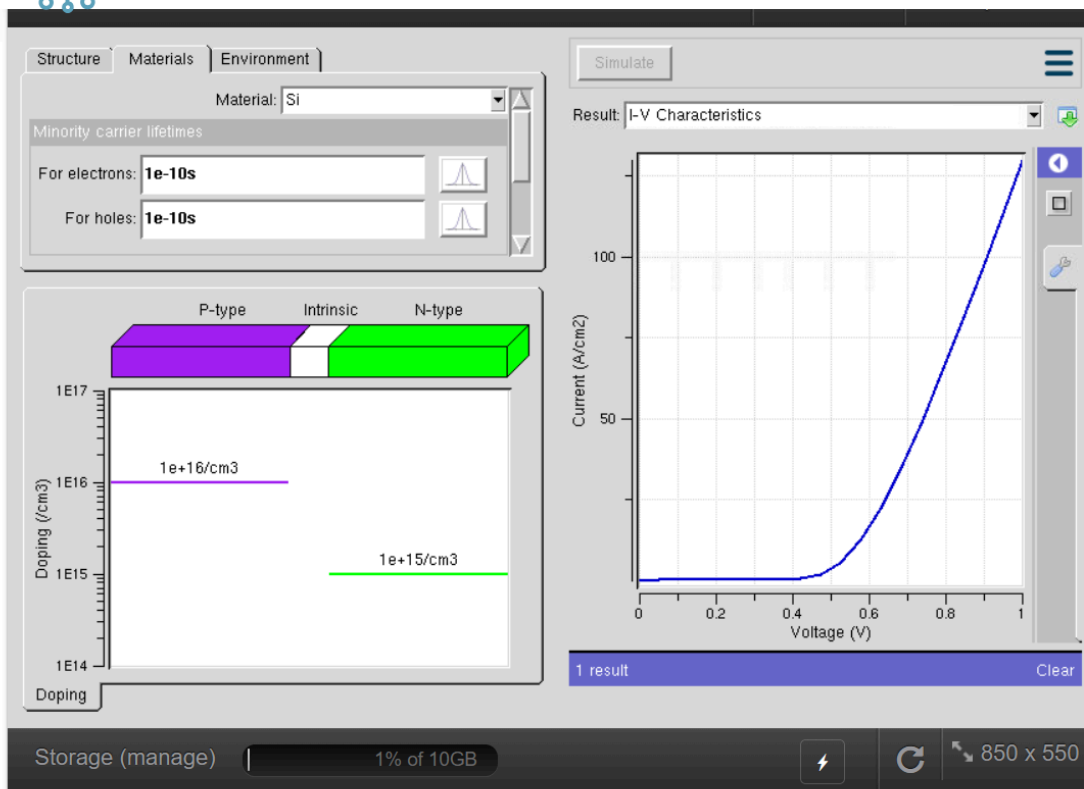
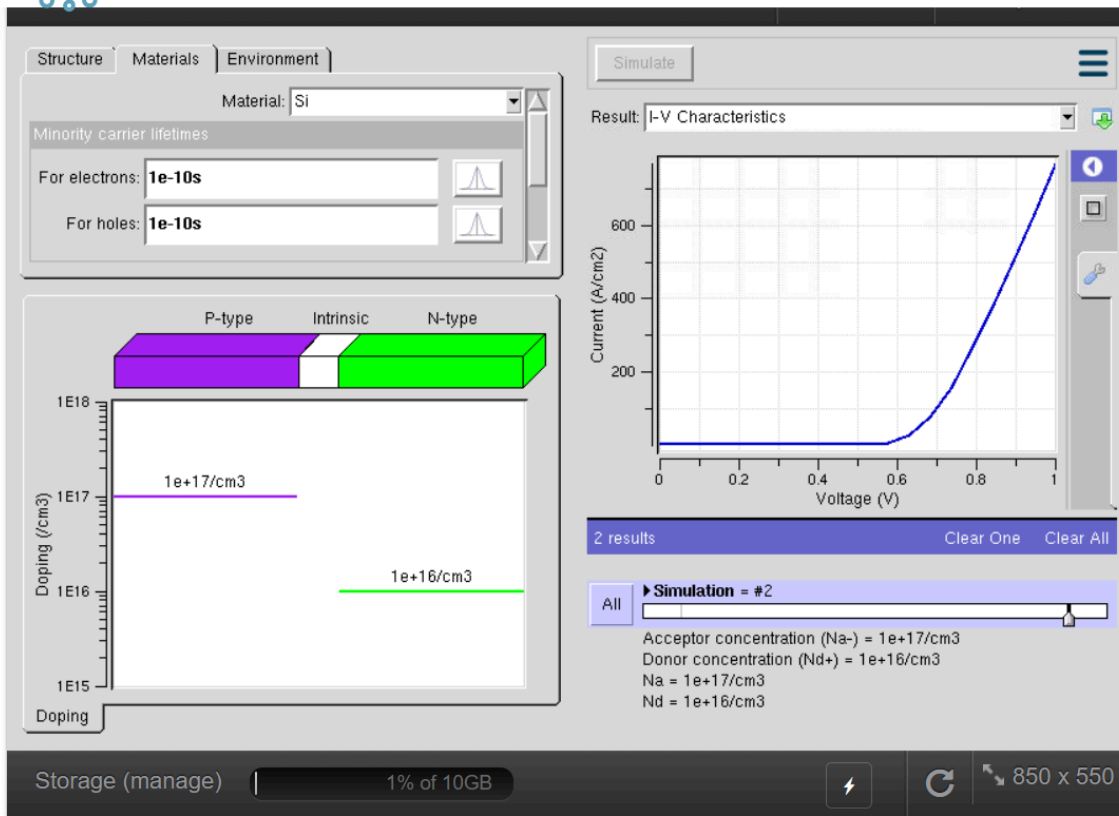
RESULTS:





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Observations:

When doping is increased, the forward current becomes much higher because the depletion width becomes narrow.

Germanium diode turns ON at lower voltage (~ 0.3 V) compared to Silicon (~ 0.7 V).

For the same doping, Ge shows higher forward current and much higher reverse leakage.

This is because Ge has lower bandgap and higher intrinsic carrier concentration.

Q2. Simulate the C–V characteristics of Metal Oxide Semiconductor for low and high frequency using nanoHUB simulation tool.

Objectives:

- Study the C-V characteristics of n-MOS capacitor for low and high frequency.
- Study the C-V characteristics of p-MOS capacitor for low and high frequency.
- Study the effect of metal work function on C-V characteristics.
- Study the effect of doping concentration on C-V characteristics.

Procedure:n-MOS capacitor – low and high frequency C–V

Select MOSCap under resources .

Enter substrate doping (Na) as , e.g. $1 \times 10^{15} \text{ cm}^{-3}$.

The work function of the gate electrode to 4.05eV.

In environment section set the low and high frequencies.

Set gate voltage sweep range, e.g. from -3 V to $+5$ V.

Run the simulation .

Procedure:p-MOS capacitor – low and high frequency C–V

Select MOSCap under resources .

Enter substrate doping (Nd) as , e.g. $1 \times 10^{15} \text{ cm}^{-3}$.

The work function of the gate electrode to 4.05eV.

In environment section set the low and high frequencies.

Set gate voltage sweep range, e.g. from -3 V to $+5$ V.

Run the simulation .

To change the frequencies in the environment section set the low and high

frequencies.Frequencies are observed for 10Hz(low frequency) and 1e+06Hz(high frequency)

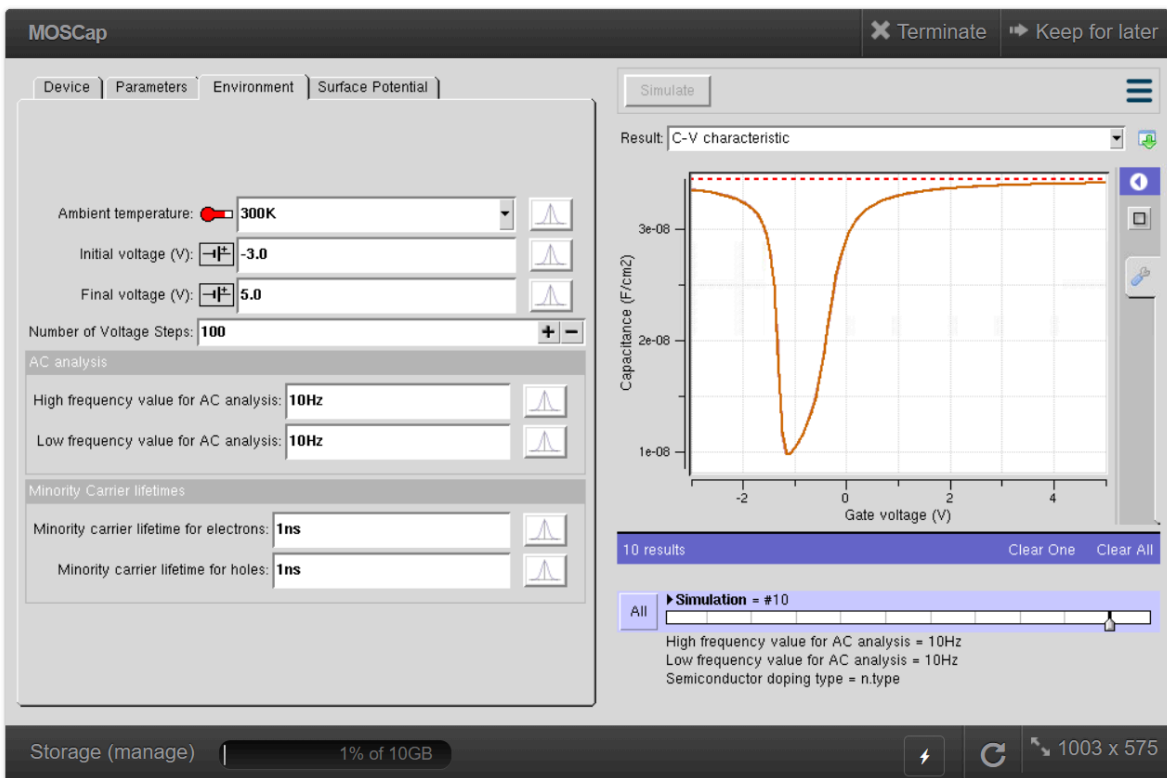
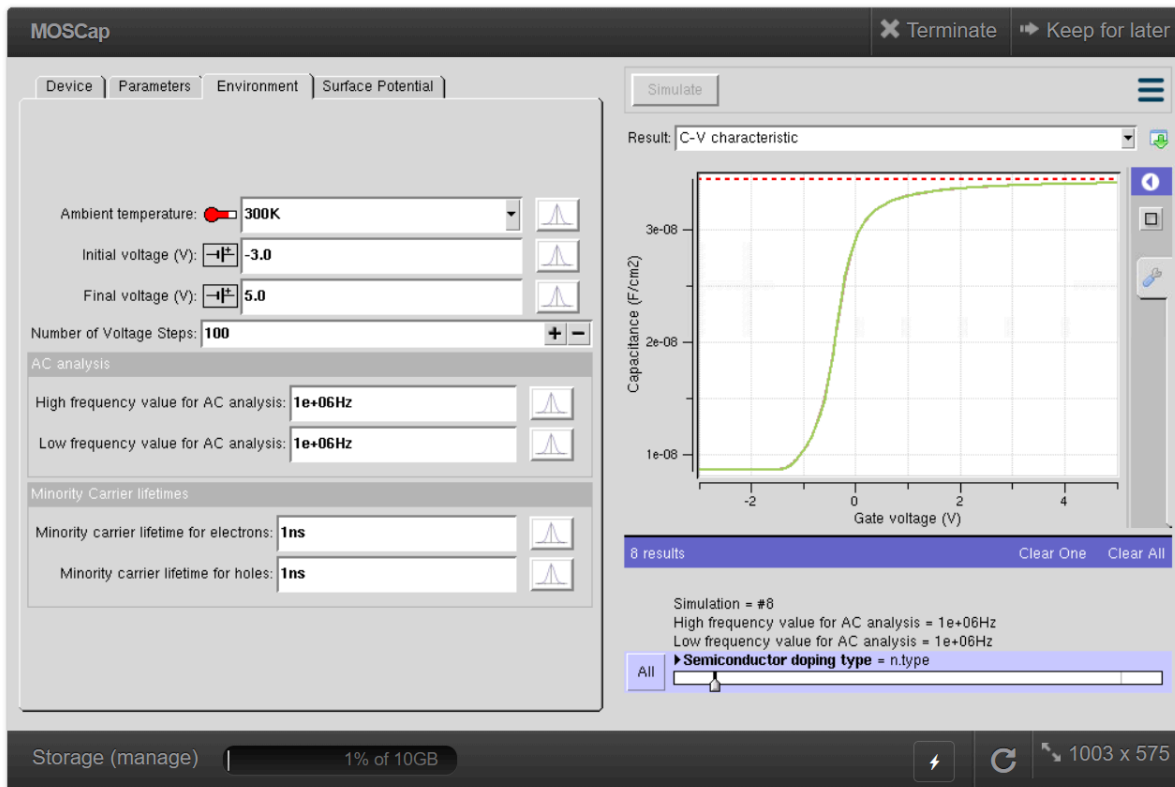
The doping concentrations can be changed by clicking the value of the

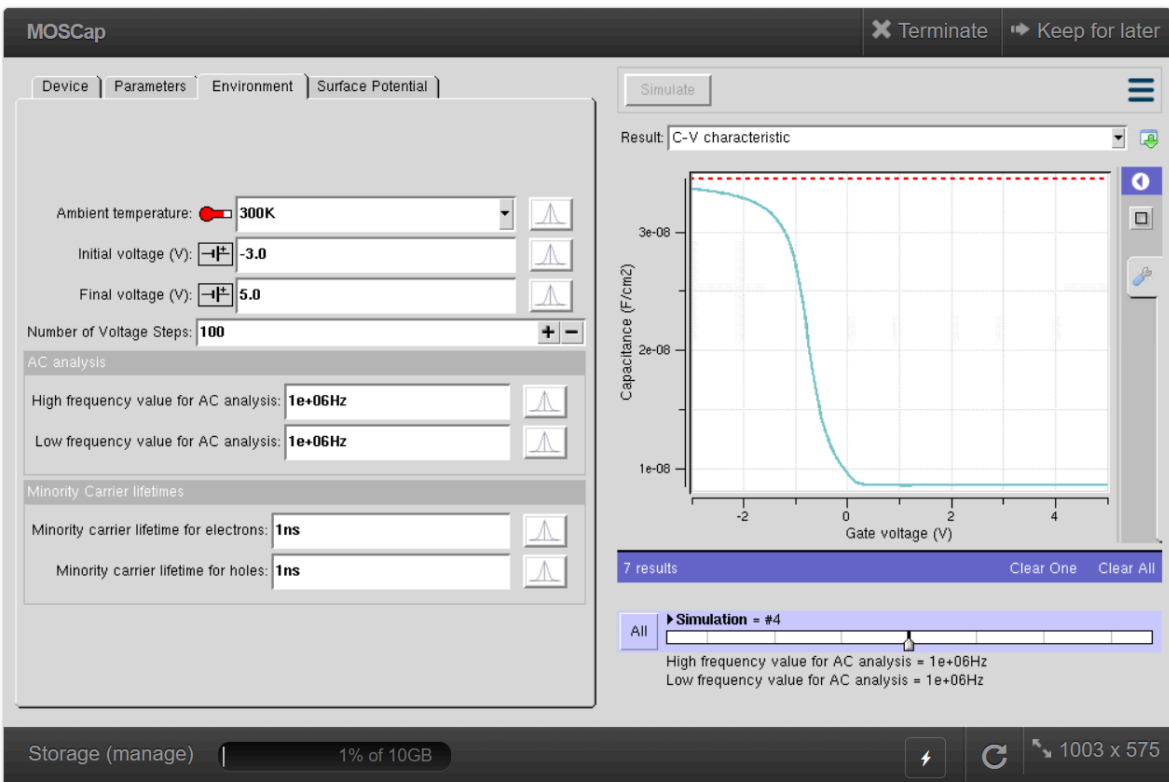
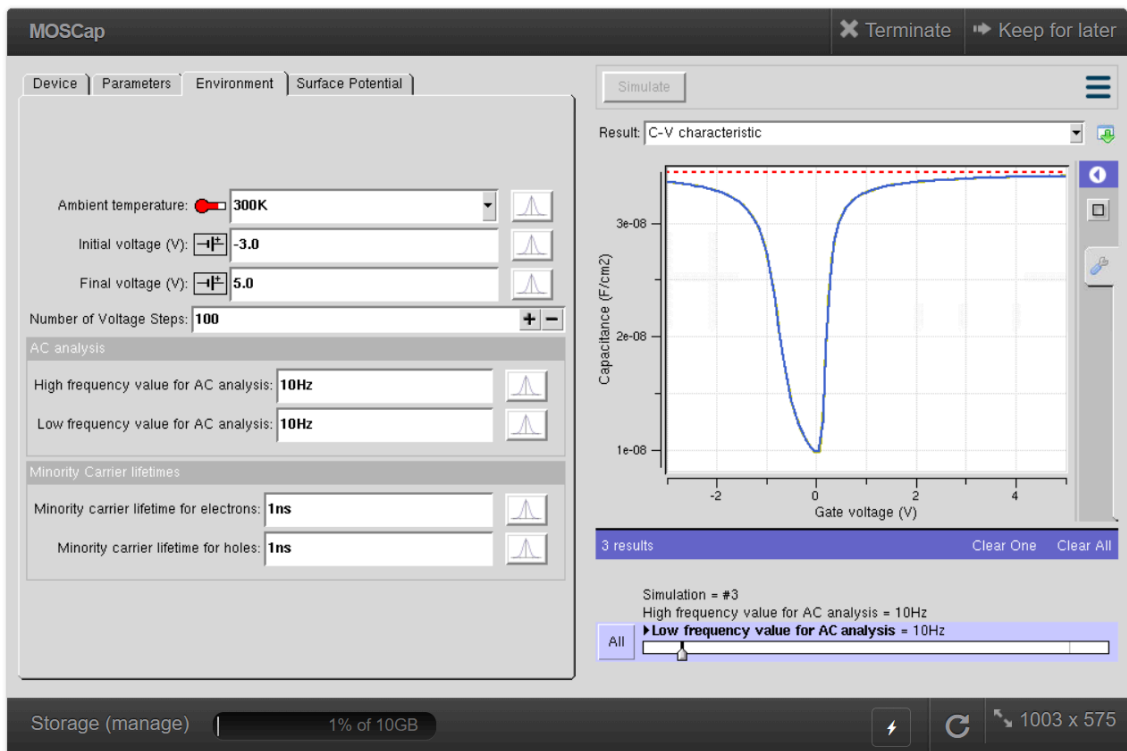
concentration.Concentration is changed from 1×10^{15} to 1×10^{16} .

The metal work function is changed from 4.05eV to 5eV.

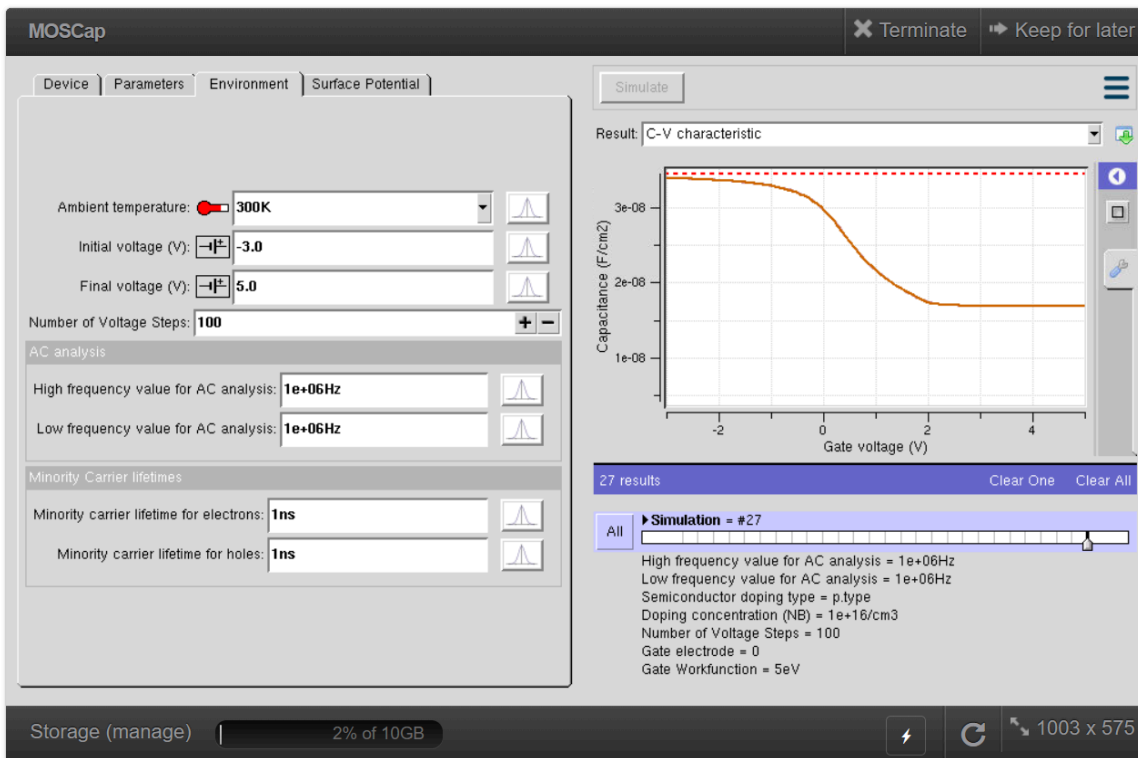
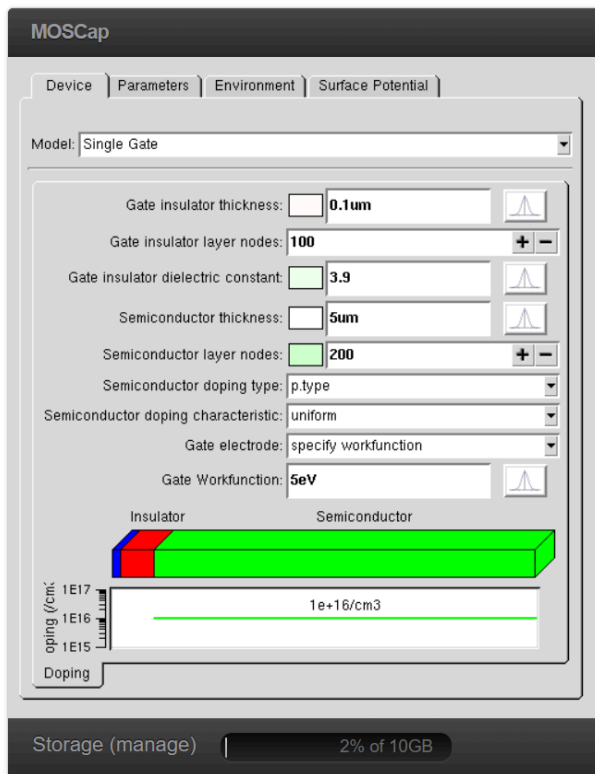
RESULTS:

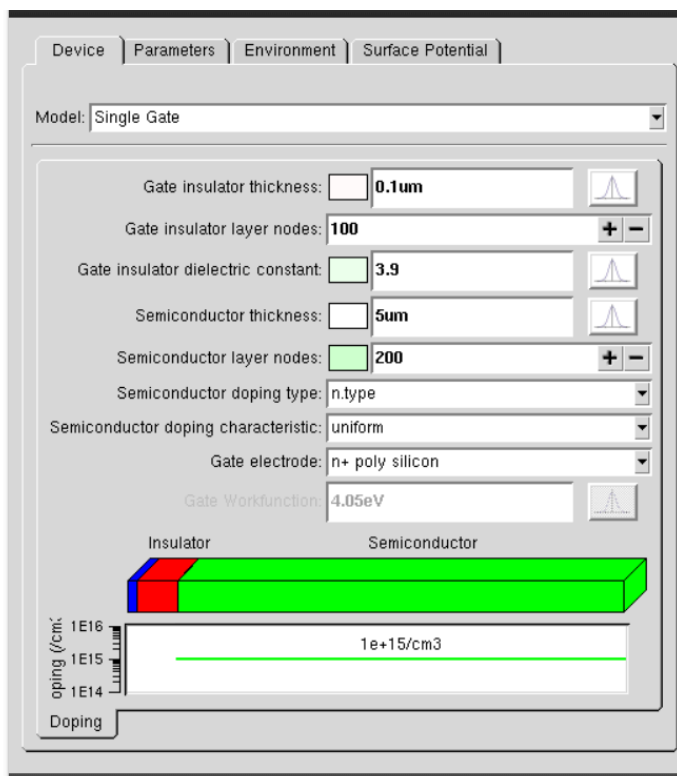
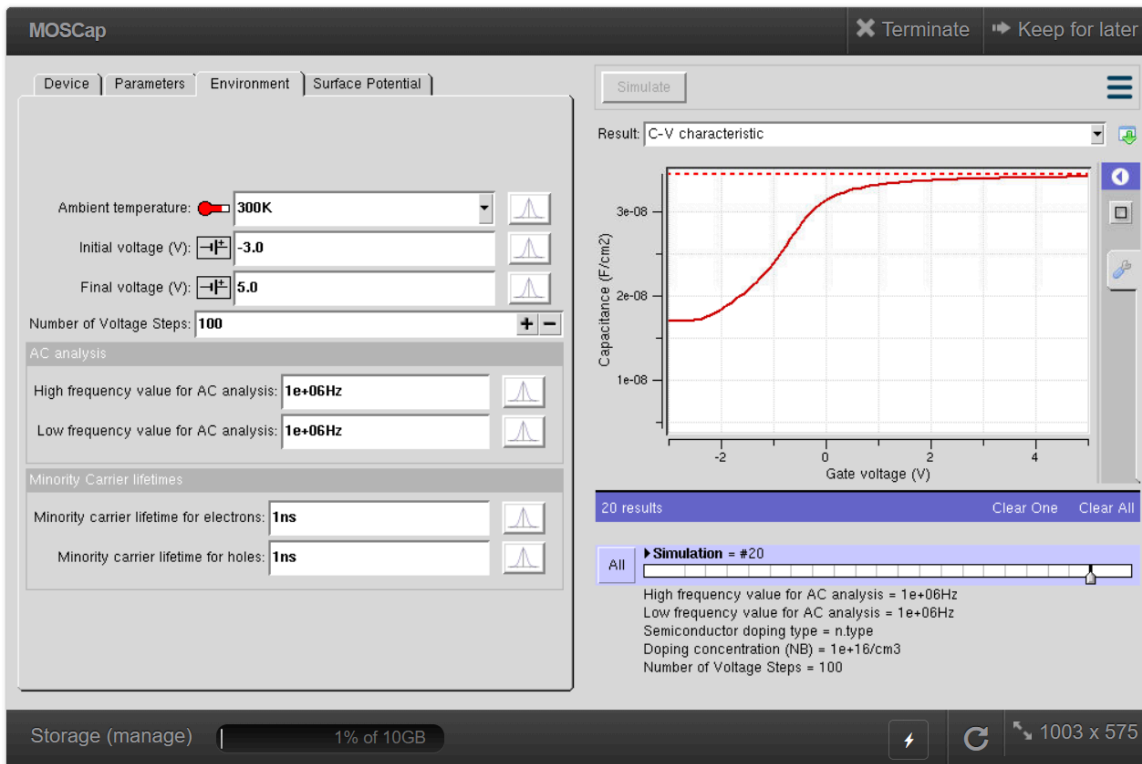
1)C-V characteristics of pmos and nmos for low and high frequencies:



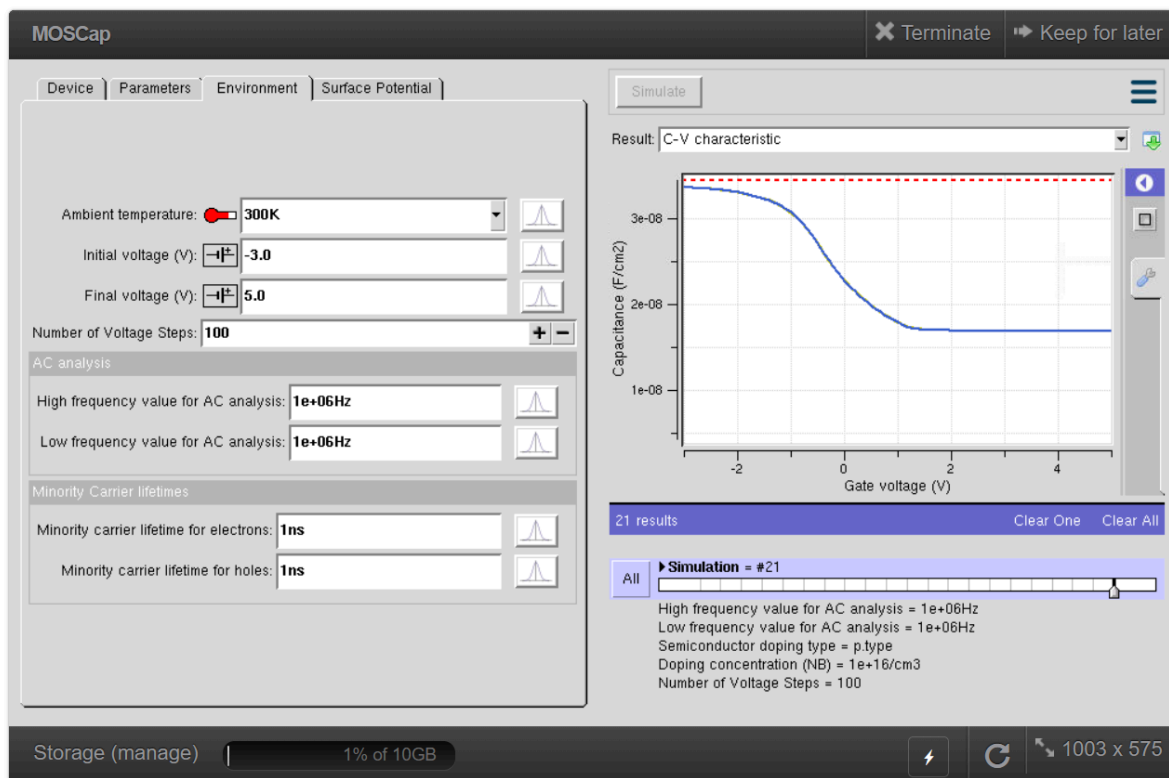
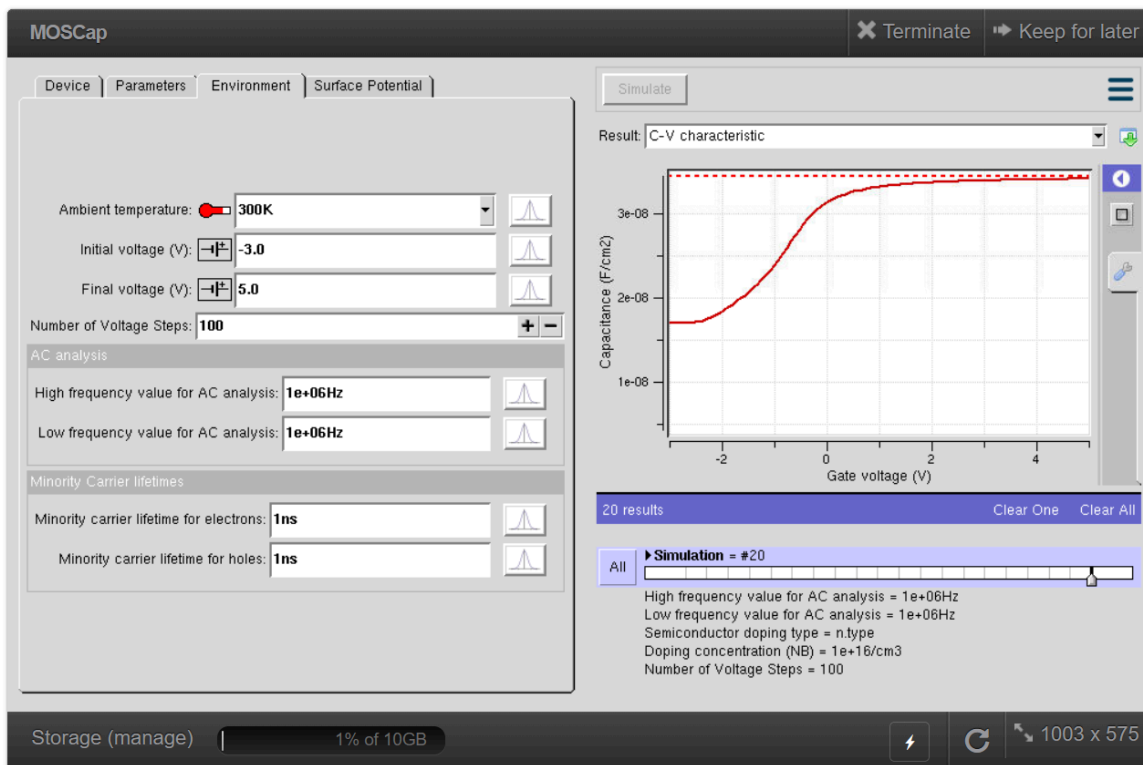


Effect of metal work function:





Effect of doping concentration



Observations:

Inversion capacitance decreases with increase in frequency. As frequency at time are inversely related at high frequency time for electrons to reach near surface reduces so capacitance decreases.

Minimum capacitance increases and threshold voltage increases.

A higher work function increases the flat-band voltage.

Q3) Simulate n-channel Metal Oxide Semiconductor Field Effect Transistor

(nMOSFET), p-channel Metal

Oxide Semiconductor Field Effect Transistor (pMOSFET) and simulate the transfer (ID–VGS)

characteristics using nanoHUB simulation tool.

Objectives:

- Study the ID–VGS characteristics of n-MOSFET for different values of VDS.
- Study the effect of doping concentration, oxide thickness on the ID–VGS characteristics.

Procedure:

Select MOSFet under resources .

Select device type = n-MOSFET.

Set channel length and width to default values

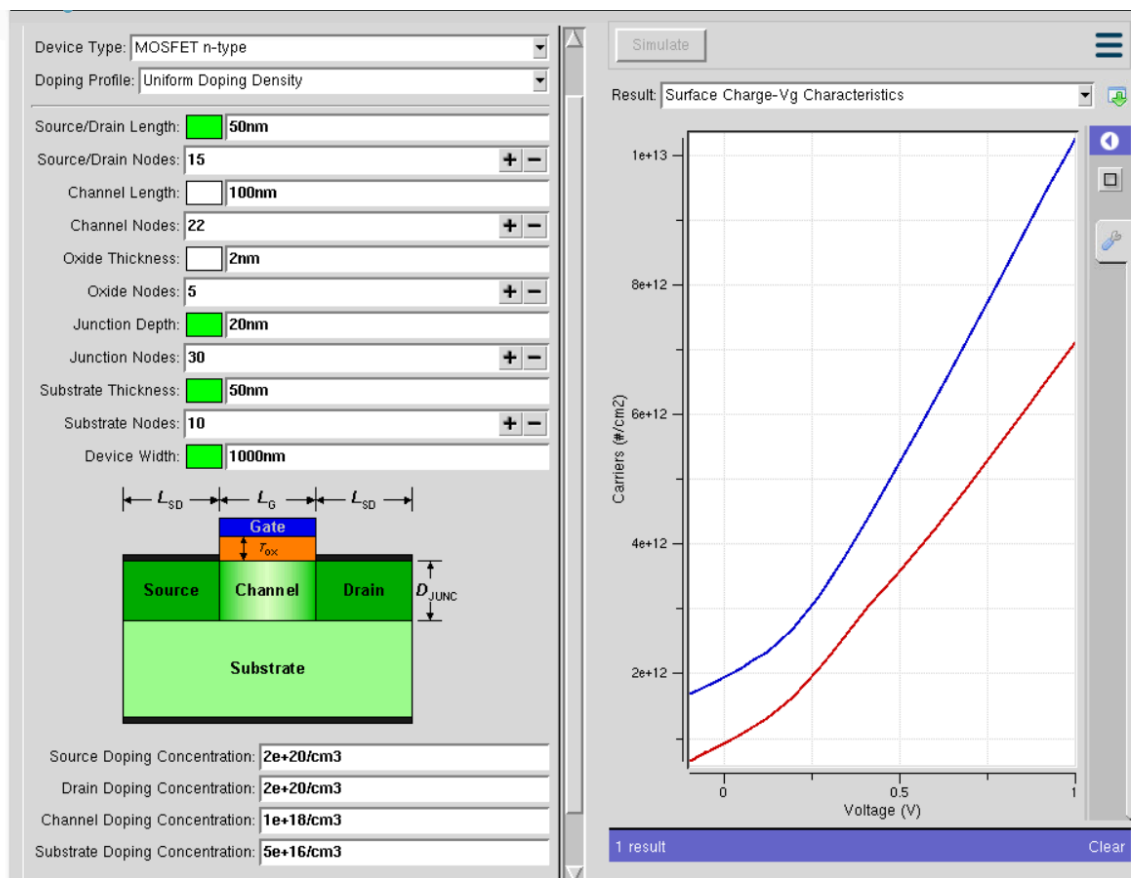
Set oxide thickness (tox), in structural properties (2nm) and change it to 4nm later.

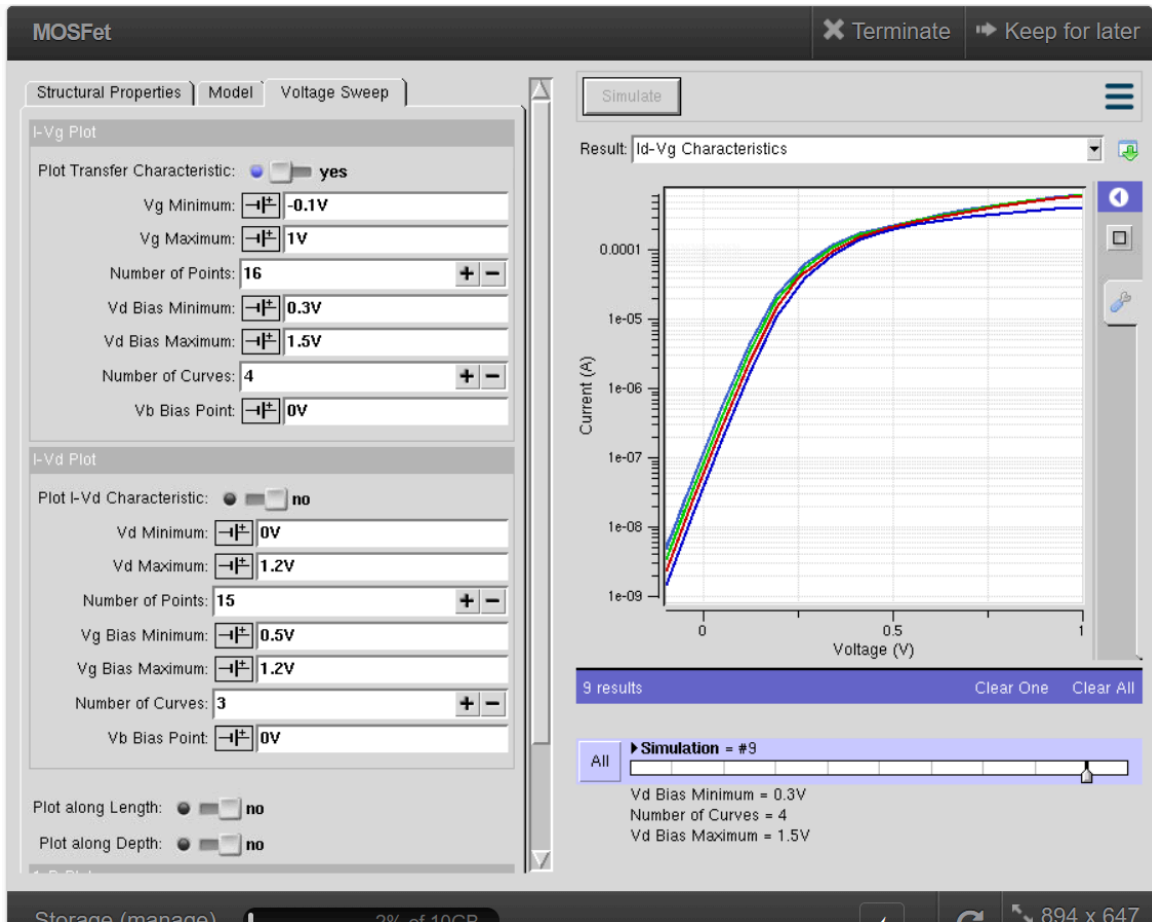
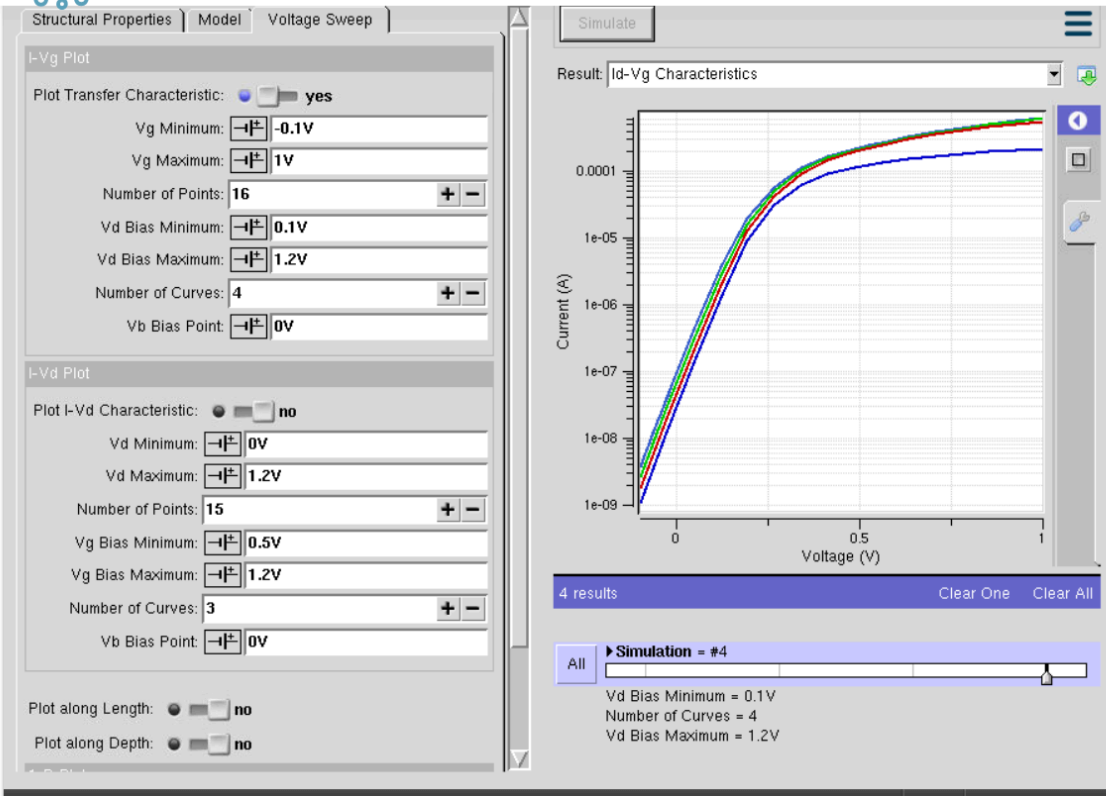
Set source and drain doping as 2e+18 and change to 2e+20 later

Choosing the drain as sweep voltage and changing the and varying vgs we observe id-vds for different vgs

Choosing the gate as sweep voltage and changing and varying the vds we observe id -vgs for different vds

Results:





Structural Properties | Model | Voltage Sweep

I-Vg Plot

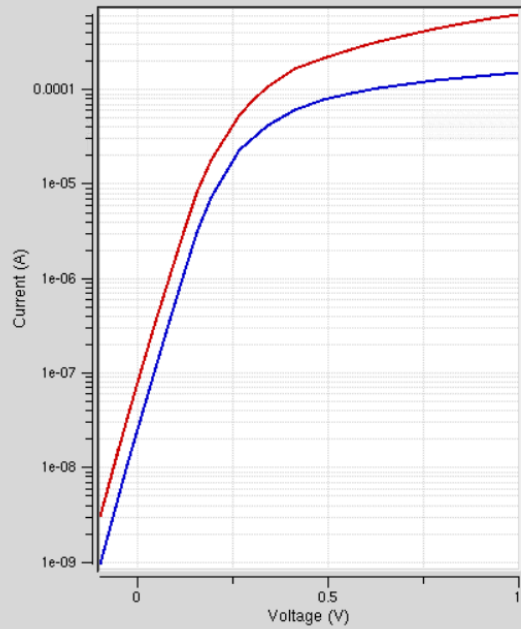
Plot Transfer Characteristic: ☒ yesVg Minimum: -0.1VVg Maximum: 1VNumber of Points: 16 Vd Bias Minimum: 0.05VVd Bias Maximum: 1VNumber of Curves: 2 Vb Bias Point: 0V

I-Vd Plot

Plot I-Vd Characteristic: ☐ ☒ noVd Minimum: 0VVd Maximum: 1.2VNumber of Points: 15 Vg Bias Minimum: 0.5VVg Bias Maximum: 1.2VNumber of Curves: 3 Vb Bias Point: 0VPlot along Length: ☐ ☒ noPlot along Depth: ☐ ☒ no

Simulate

Result: Id-Vg Characteristics



1 result

Clear

Changed Parameters

Device Type:	MOSFET n-type
Doping Profile:	Uniform Doping Density

Source/Drain Length:	50nm		
Source/Drain Nodes:	15	+	-
Channel Length:	100nm		
Channel Nodes:	22	+	-
Oxide Thickness:	4nm		
Oxide Nodes:	5	+	-
Junction Depth:	20nm		
Junction Nodes:	30	+	-
Substrate Thickness:	50nm		
Substrate Nodes:	10	+	-
Device Width:	1000nm		

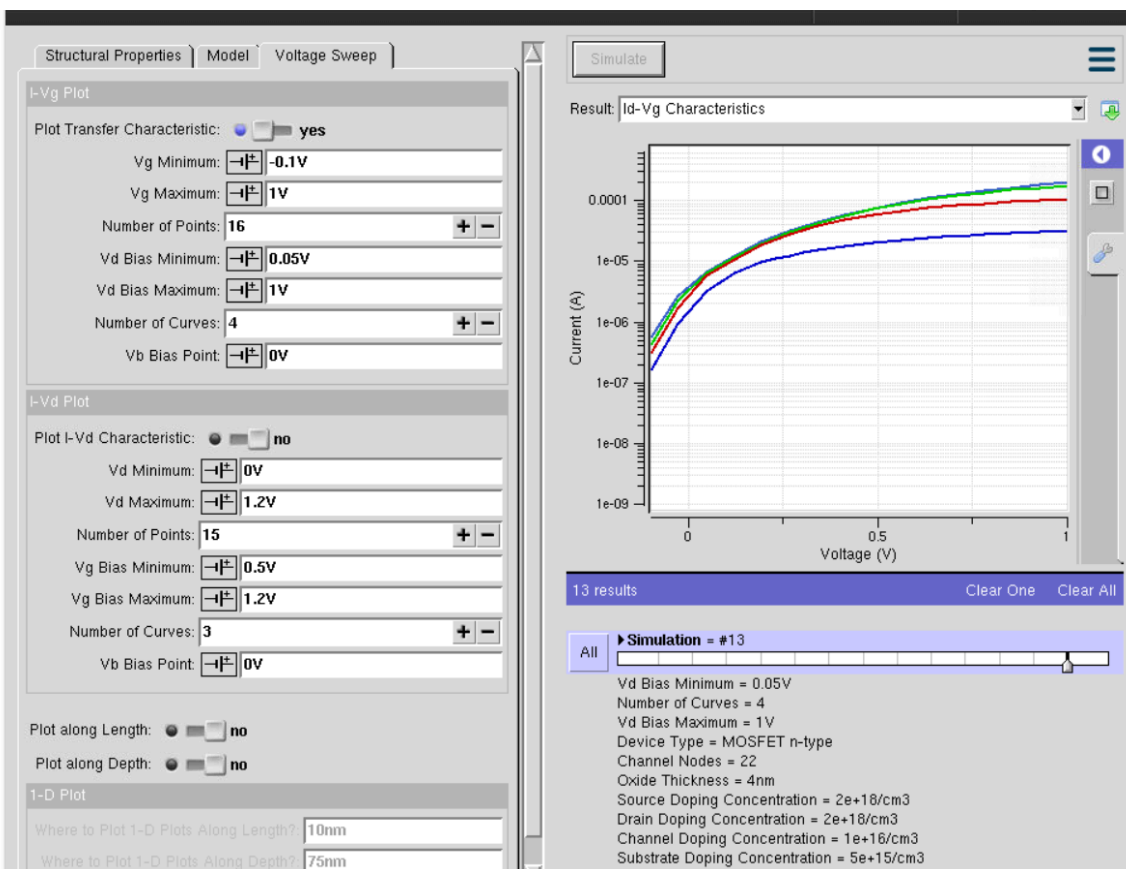
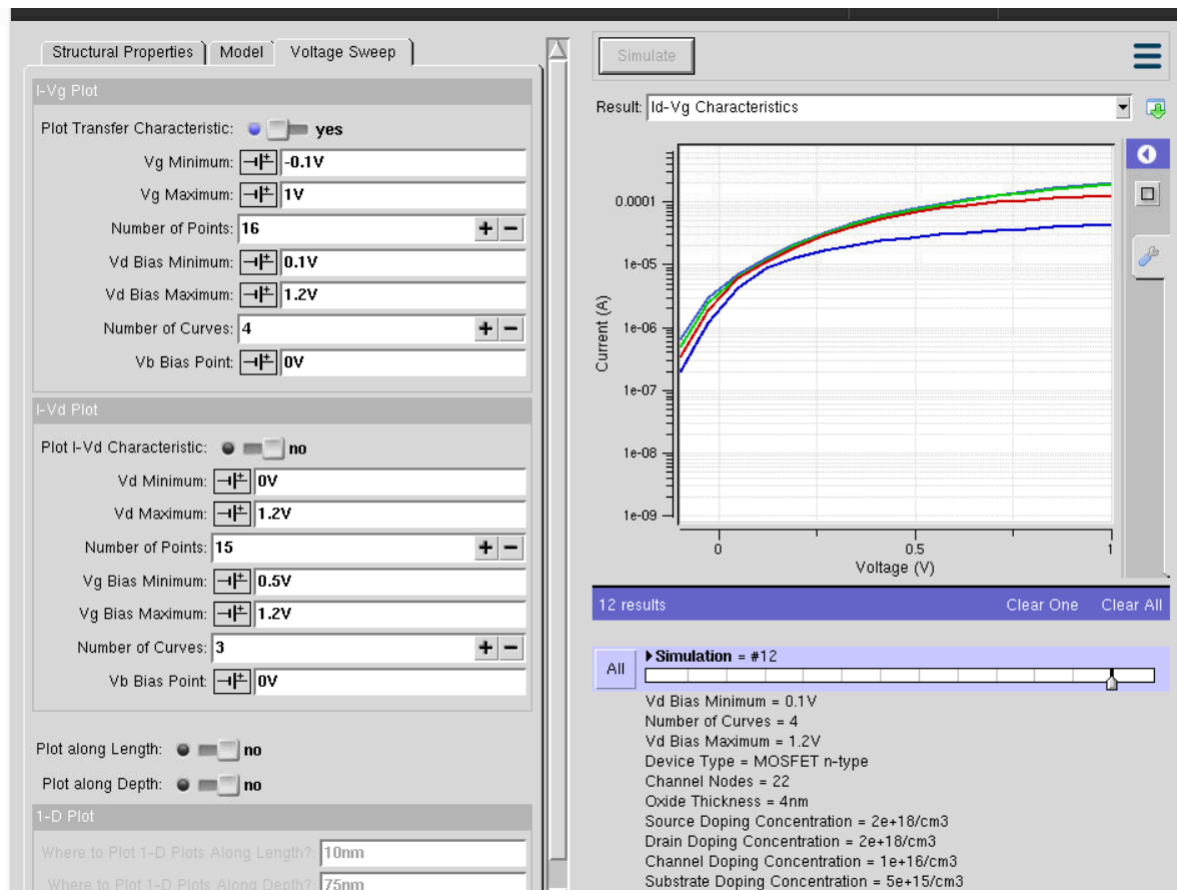
Diagram illustrating the MOSFET structure and parameters:

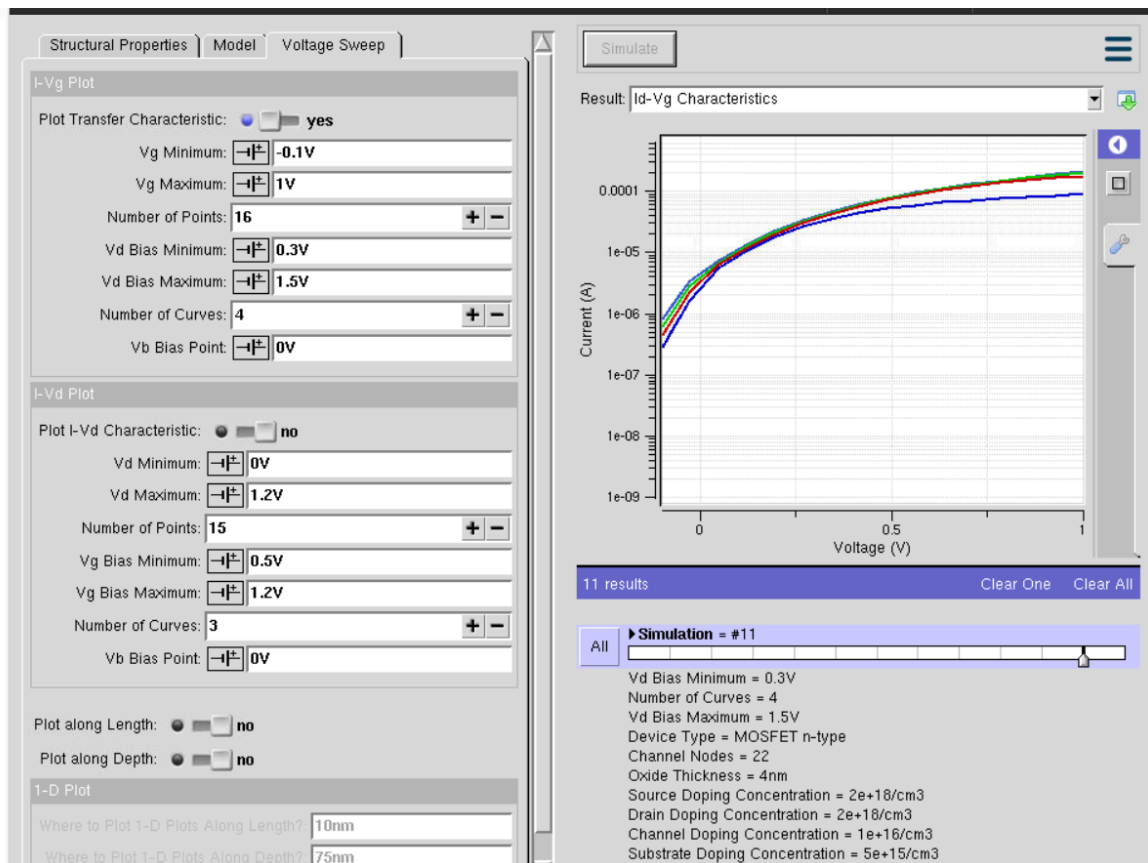
- Source/Drain Length: 50nm
- Source/Drain Nodes: 15
- Channel Length: 100nm
- Channel Nodes: 22
- Oxide Thickness: 4nm
- Oxide Nodes: 5
- Junction Depth: 20nm
- Junction Nodes: 30
- Substrate Thickness: 50nm
- Substrate Nodes: 10
- Device Width: 1000nm

Diagram labels:

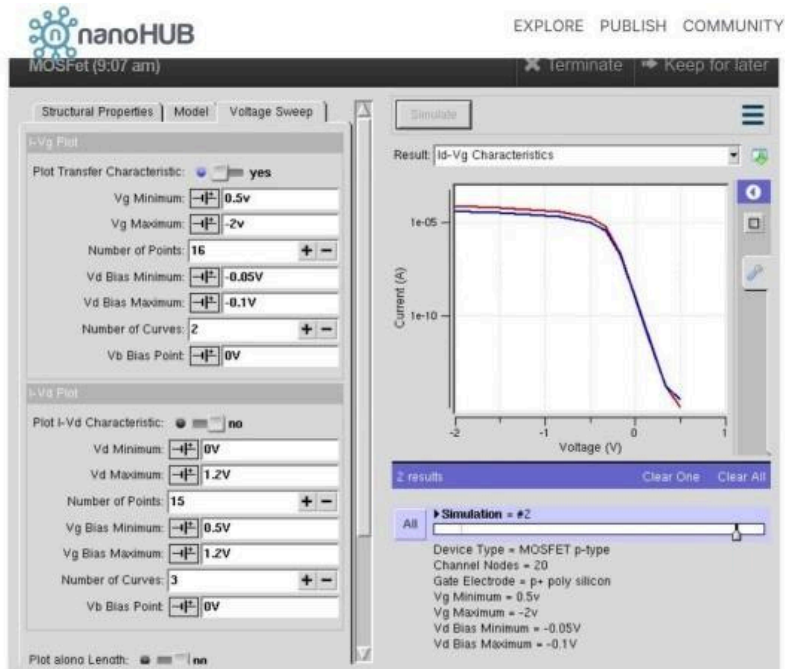
- L_{SD} (Source/Drain Length)
- L_G (Channel Length)
- T_{ox} (Oxide Thickness)
- D_{JUNC} (Junction Depth)
- Source, Channel, Drain, Substrate, Gate

Source Doping Concentration:	2e+18/cm3
Drain Doping Concentration:	2e+18/cm3
Channel Doping Concentration:	1e+16/cm3
Substrate Doping Concentration:	5e+15/cm3





PMOSFet



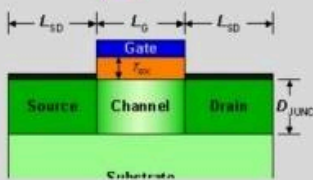
MOSFet (9:07 am)

Terminate Keep for later

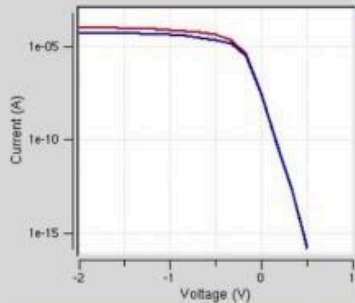
Structural Properties Model Voltage Sweep

Device Type: MOSFET p-type
Doping Profile: Uniform Doping Density

Source/Drain Length: 50nm
Source/Drain Nodes: 15
Channel Length: 100nm
Channel Nodes: 20
Oxide Thickness: 1nm
Oxide Nodes: 5
Junction Depth: 20nm
Junction Nodes: 30
Substrate Thickness: 50nm
Substrate Nodes: 10
Device Width: 1000nm



Result: Id-Vg Characteristics



3 results Clear One Clear All

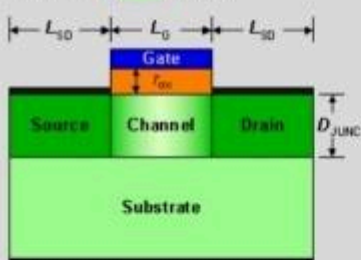
Simulation = #3
Device Type = MOSFET p-type
Channel Nodes = 20
Gate Electrode = p+ poly silicon
Vg Minimum = 0.5v
Vg Maximum = -2v
Vd Bias Minimum = -0.05V
Vd Bias Maximum = -0.1V

Oxide Thickness = 1nm

MOSFet (9:07 am)

Terminate Keep for later

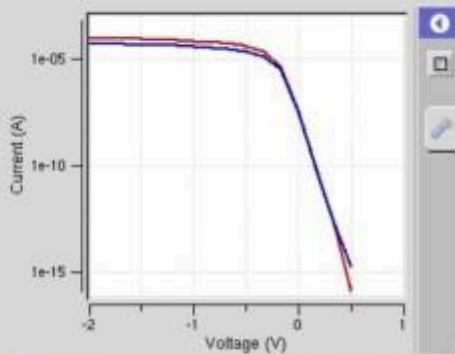
Oxide Thickness: 1nm
Oxide Nodes: 5
Junction Depth: 20nm
Junction Nodes: 30
Substrate Thickness: 50nm
Substrate Nodes: 10
Device Width: 1000nm



Source Doping Concentration: $1e+20/cm^3$
Drain Doping Concentration: $1e+20/cm^3$
Channel Doping Concentration: $1e+18/cm^3$
Substrate Doping Concentration: $3e+16/cm^3$

Simulate

Result: Id-Vg Characteristics



4 results Clear One Clear All

Simulation = #4

Device Type = MOSFET p-type
Channel Nodes = 20
Gate Electrode = p+ poly silicon
Vg Minimum = 0.5v
Vg Maximum = -2v
Vd Bias Minimum = -0.05V
Vd Bias Maximum = -0.1V
Oxide Thickness = 1nm
Source Doping Concentration = $1e+20/cm^3$
Drain Doping Concentration = $1e+20/cm^3$
Channel Doping Concentration = $1e+18/cm^3$
Substrate Doping Concentration = $3e+16/cm^3$

Observations:

n-Mosfet:

I_D increases when V_{GS} increases because a stronger electric field forms more electrons in the channel, creating a wider and more conductive n-type path from source to drain.

As V_{GS} rises above threshold, the inversion layer becomes thicker, so the channel resistance drops and current flows more easily.

Higher V_{GS} pulls electrons closer to the oxide interface, giving better gate control and increasing carrier mobility toward the drain.

A thinner oxide increases gate capacitance, so the gate can attract more electrons for the same voltage, leading to higher I_D .

p-Mosfet:

As V_{GS} becomes less negative, the gate electric field becomes weaker, so fewer holes are attracted to form the channel, reducing current.

When the negative gate voltage reduces, the inversion layer becomes thinner, increasing channel resistance and dropping the drain current.

Holes have lower mobility than electrons, so pMOS current naturally reduces faster compared to nMOS when gate control weakens.

Thinner oxide increases hole attraction (stronger field), so I_D improves because the channel forms more efficiently.